

Day 65 – Dynamic Routing with RIP Version 2

Overview

Day 65 introduced dynamic routing using Routing Information Protocol Version 2 (RIP v2). Unlike static routing, where routes must be manually configured on every router, RIP allows routers to automatically exchange routing information and learn about remote networks.

In this lab, a multi-router topology was built consisting of three LANs connected through three routers. RIP v2 was configured on all routers to automatically advertise and learn routes.

This lab demonstrated how routing protocols simplify network management in larger networks.

Objectives

By the end of this lab, I was able to:

- Configure router interfaces and LAN addressing
 - Build a multi-router topology
 - Configure point-to-point router links using /30 networks
 - Configure RIP version 2
 - Verify dynamic route learning
 - Troubleshoot connectivity issues
 - Test communication between remote LANs
-

Network Topology

Logical Design

LAN10 LAN20 LAN30

PC0 PC1 PC2
|||
SW0 SW1 SW2
|||
R0 ----- R1 ----- R2

Addressing Plan

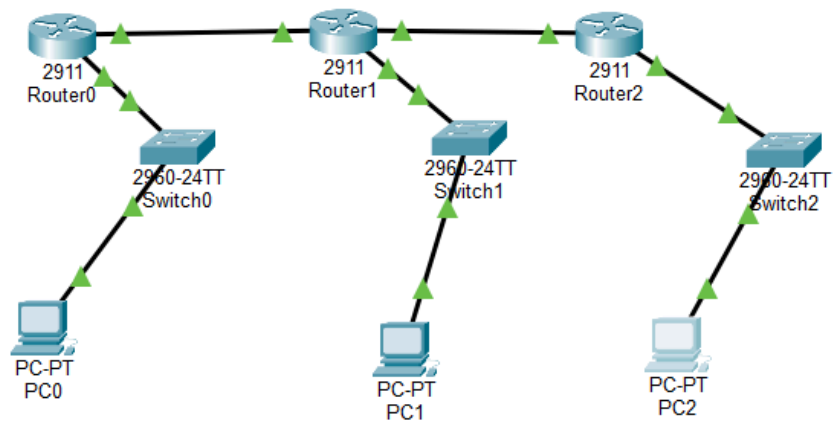
Device	Interface	Address
Router0	G0/0	192.168.10.1/26
Router0	G0/1	10.0.0.1/30
Router1	G0/0	192.168.20.1/26
Router1	G0/1	10.0.0.2/30
Router1	G0/2	10.0.0.5/30
Router2	G0/0	192.168.30.1/26
Router2	G0/1	10.0.0.6/30

PC Configuration

Device	IP Address	Default Gateway
PC0	192.168.10.10	192.168.10.1
PC1	192.168.20.10	192.168.20.1
PC2	192.168.30.10	192.168.30.1

Screenshot #1 – Complete Topology

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- Three routers
 - Three switches
 - Three PCs
 - Router-to-router links
 - All links green
-

Router Interface Configuration

Router0

```
interface g0/0
ip address 192.168.10.1 255.255.255.192
no shutdown
```

```
interface g0/1
ip address 10.0.0.1 255.255.255.252
no shutdown
```

Router1

```
interface g0/0
ip address 192.168.20.1 255.255.255.192
no shutdown
```

```
interface g0/1
ip address 10.0.0.2 255.255.255.252
no shutdown
```

```
interface g0/2
ip address 10.0.0.5 255.255.255.252
no shutdown
```

Router2

```
interface g0/0
ip address 192.168.30.1 255.255.255.192
no shutdown
```

```
interface g0/1
ip address 10.0.0.6 255.255.255.252
no shutdown
```

Screenshot #2 – Router0 Interface Verification

show ip interface brief

```
Router#show ip interface brief
Interface          IP-Address      OK? Method Status              Protocol
GigabitEthernet0/0 192.168.10.1    YES manual up                  up
GigabitEthernet0/1 10.0.0.1        YES manual up                  up
GigabitEthernet0/2 unassigned      YES unset  down                down
Vlan1              unassigned      YES unset  administratively down down
Router#
```

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- G0/0 = 192.168.10.1
 - G0/1 = 10.0.0.1
 - Status = up/up
-

Screenshot #3 – Router1 Interface Verification

```
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  192.168.20.1    YES manual up          up
GigabitEthernet0/1  10.0.0.2        YES manual up          up
GigabitEthernet0/2  10.0.0.5        YES manual up          up
Vlan1           unassigned      YES unset  administratively down down
```

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Highlight

- G0/0 = 192.168.20.1
 - G0/1 = 10.0.0.2
 - G0/2 = 10.0.0.5
 - Status = up/up
-

Screenshot #4 – Router2 Interface Verification

```
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  192.168.30.1    YES manual up          up
GigabitEthernet0/1  10.0.0.6        YES manual up          up
GigabitEthernet0/2  unassigned      YES unset  down        down
Vlan1           unassigned      YES unset  administratively down down
```

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Highlight

- G0/0 = 192.168.30.1
 - G0/1 = 10.0.0.6
 - Status = up/up
-

RIP Version 2 Configuration

Router0

```
router rip
version 2
network 192.168.10.0
network 10.0.0.0
```

Router1

```
router rip
version 2
network 192.168.20.0
network 10.0.0.0
```

Router2

```
router rip
version 2
network 192.168.30.0
network 10.0.0.0
```

Routing Table Verification

After enabling RIP, each router successfully learned routes to remote networks.

Example Router0 Routing Table

```
show ip route
```

Learned routes:

R 192.168.20.0/24

R 192.168.30.0/24

Meaning:

- Router0 automatically learned LAN20
- Router0 automatically learned LAN30

without static routes.

Screenshot #5 – RIP Route Learning

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```
      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/1
R       10.0.0.4/30 [120/1] via 10.0.0.2, 00:00:03, GigabitEthernet0/1
      192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/26 is directly connected, GigabitEthernet0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0
R       192.168.20.0/24 [120/1] via 10.0.0.2, 00:00:03, GigabitEthernet0/1
R       192.168.30.0/24 [120/2] via 10.0.0.2, 00:00:03, GigabitEthernet0/1
```

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```
      10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.2/32 is directly connected, GigabitEthernet0/1
C       10.0.0.4/30 is directly connected, GigabitEthernet0/2
L       10.0.0.5/32 is directly connected, GigabitEthernet0/2
R       192.168.10.0/24 [120/1] via 10.0.0.1, 00:00:04, GigabitEthernet0/1
      192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/26 is directly connected, GigabitEthernet0/0
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0
R       192.168.30.0/24 [120/1] via 10.0.0.6, 00:00:04, GigabitEthernet0/2
```

Router#

```

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
R       10.0.0.0/30 [120/1] via 10.0.0.5, 00:00:18, GigabitEthernet0/1
C       10.0.0.4/30 is directly connected, GigabitEthernet0/1
L       10.0.0.6/32 is directly connected, GigabitEthernet0/1
R       192.168.10.0/24 [120/2] via 10.0.0.5, 00:00:18, GigabitEthernet0/1
R       192.168.20.0/24 [120/1] via 10.0.0.5, 00:00:18, GigabitEthernet0/1
        192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/26 is directly connected, GigabitEthernet0/0
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0
-

```

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Look for:

R 192.168.20.0

R 192.168.30.0

The letter:

R

indicates routes learned through RIP.

Network Testing

Router Connectivity

Successful tests:

ping 10.0.0.2

ping 10.0.0.5

ping 10.0.0.6

Results confirmed that router-to-router communication was functioning properly.

End-to-End LAN Testing

PC0 → PC1

ping 192.168.20.10

Successful.

PC0 → PC2

ping 192.168.30.10

Successful.

PC1 → PC2

ping 192.168.30.10

Successful.

PC2 → PC0

ping 192.168.10.10

Successful.

Screenshot #6 – PC0 Remote Ping

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```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
Reply from 192.168.20.10: bytes=32 time=1ms TTL=126
Reply from 192.168.20.10: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 1ms

C:\>
C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.10: bytes=32 time<1ms TTL=125
Reply from 192.168.30.10: bytes=32 time<1ms TTL=125
Reply from 192.168.30.10: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

Highlight

- Successful replies
- Remote network communication

Screenshot #7 – PC2 Remote Ping

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```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=125
Reply from 192.168.10.10: bytes=32 time<1ms TTL=125
Reply from 192.168.10.10: bytes=32 time<1ms TTL=125
Reply from 192.168.10.10: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
Reply from 192.168.20.10: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Highlight

- End-to-end communication
 - RIP functioning correctly
-

Troubleshooting Performed

During the lab, router-to-router pings initially failed.

Investigation revealed that the physical cabling did not match the interface addressing plan. Router interfaces had been configured correctly, but the cabling connected different interfaces than expected.

After verifying interface assignments and correcting the cables:

- Router-to-router pings succeeded

- RIP exchanged routes successfully
- End-to-end communication worked

This troubleshooting process reinforced the importance of matching logical network design with physical topology.

What I Learned

- Dynamic routing reduces manual configuration.
 - RIP routers automatically exchange routing information.
 - The `show ip route` command verifies learned routes.
 - The `R` route code indicates RIP-learned routes.
 - Point-to-point networks commonly use /30 addressing.
 - Correct cabling is just as important as correct configuration.
 - Routing issues should be troubleshot layer by layer.
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Reflection

Day 65 was my first complete multi-router dynamic routing lab. While rebuilding the topology, I encountered a routing failure that was ultimately traced to an interface cabling mismatch. Troubleshooting and correcting this issue helped reinforce how physical topology and logical addressing work together. After enabling RIP v2, all routers successfully exchanged routes and remote LAN communication worked without using static routes. This lab helped me better understand how enterprise networks automatically share routing information and prepared me for Day 66, where DHCP will automate host IP addressing across the same routed network.
